

SDLC Model Selection For Hospital Management

Comparison Matrix:

A comparative analysis of different SDLC models was conducted based on selected project criteria. Each criterion was assigned a priority according to its importance to the system.

The models were then evaluated against these criteria, and scores were calculated accordingly.

Criteria	Priority	Waterfall	V-Shape	Iterative	Spiral	Agile	Prototype
Requirement Stability	3	Yes	Yes	No	No	No	No
Usability & User Feedback	5	No	No	Yes	Yes	Yes	Yes
Dependability & Security	5	No	Yes	Yes	Yes	No	No
Accommodates Changes	4	No	No	Yes	Yes	Yes	Yes
Modular System Design	4	No	No	Yes	Yes	Yes	No

Cost-Effectiveness	4	No	No	Yes	No	Yes	Yes
Total Score (Out of 25)	--	3	8	22	18	17	13

Result and Justification:

The **Iterative Model** achieves the **highest score (22)**, making it the definitive choice for the Hospital Patient-Doctor Management System.

The core strength of the Iterative approach is its development in repeated cycles , which aligns perfectly with your requirement for a modular system design. By using this model, your team can develop and deploy core modules first—such as the Smart Appointment System —and test them with actual receptionists and patients to ensure a highly user-friendly interface.

Because it is highly suitable when requirements evolve, you can refine the UI/UX based on patient and doctor feedback before heavily investing in secondary features. While the Spiral model also handles evolving requirements effectively, it is designed for extremely large, high-risk projects and carries a high analytical cost. The Iterative model delivers the necessary flexibility and rigorous testing per cycle while remaining cost-effective enough to respect your project's budget and time constraints.